AirfoilTools

2D aerodynamic analysis of airfoils

User Manual

Version 3.1 — June 2026

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AirfoilTools — User Manual, version 3.1
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This document describes the use of the **AirfoilTools** library and its graphical interface for the 2D aerodynamic analysis of airfoils through coupling with the *XFoil* solver.

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Overview

1.1 About AirfoilTools

AirfoilTools is a Python library of tools dedicated to the 2D aerodynamic design and analysis of airfoils, propellers and other profiled sections. It relies on Bézier curves of arbitrary degree and multi-segment splines for geometric modeling, and integrates a coupling with the *XFoil* aerodynamic solver for performance analysis.

The application combines:

- a **geometric engine** based on Bézier curves of arbitrary degree and multi-segment splines (BezierSpline) with configurable continuities (C^0, C^1, C^2);
- a **coupling with XFoil**, the reference aerodynamic solver for viscous subsonic 2D flows;
- an **interactive graphical interface** allowing airfoil editing, simulation control and comparative visualization of results;
- a **tracing mode**: loading a background image (photo, scan, drawing) to reconstruct an airfoil by overlaying the control points onto the silhouette, all of which can be saved in a **project** file (`.aftproj`);
- a **NACA airfoil generator** (4 and 5 digits) directly from the interface;
- a **flap module**: generation of a profile deflected about a hinge axis, which can be simulated and compared to the original airfoil;
- **geometric comparison tools**: deviation between two airfoils (vertical or normal) and curvature plotting.

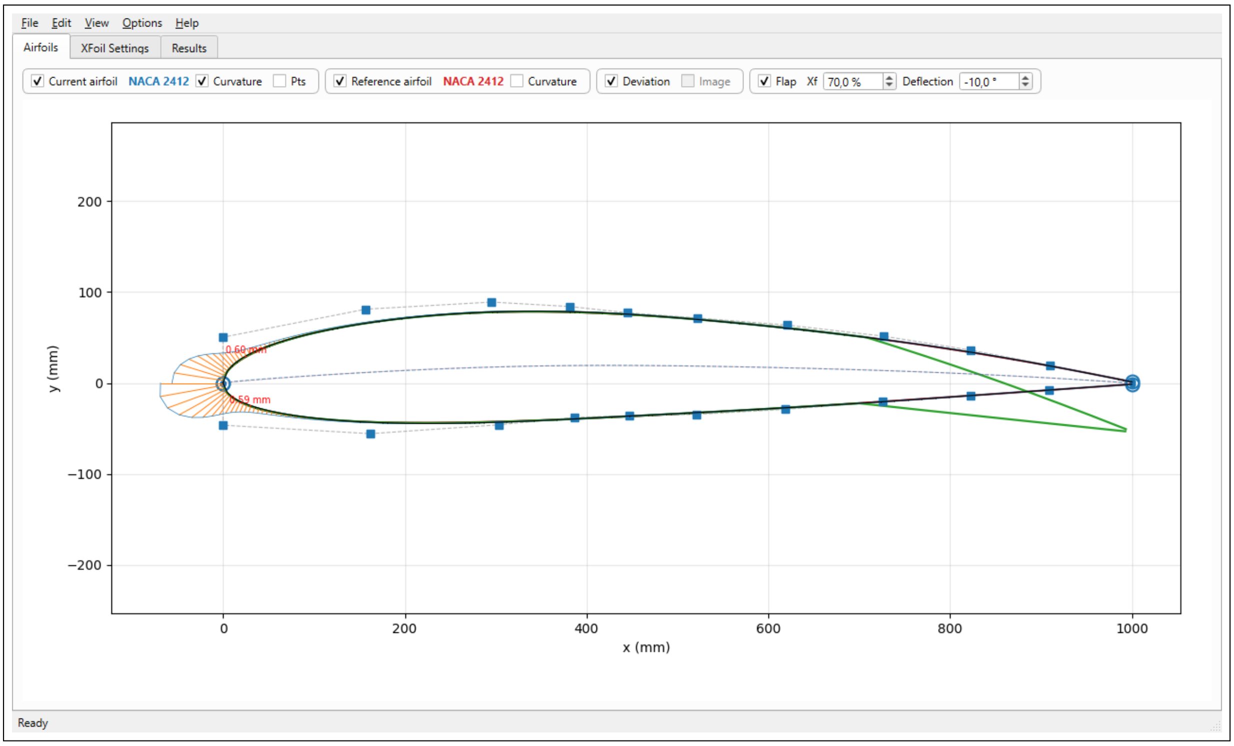

Figure 1.1: General view of the application at startup. The current airfoil (NACA 2412, blue, converted to Bézier) is overlaid on the reference airfoil (NACA 2412, red); the airfoil with deflected flap (green) as well as the curvature and the deviation are displayed by default.

1.2 Who this manual is for

This manual is intended for the **users** of the graphical application. It covers all the features accessible from the interface, from loading an airfoil to the comparative analysis of aerodynamic polars.

Developers interested in the Python API of the library will find the technical documentation in the source code (Sphinx-format docstrings) and in the generated documentation (`docs/` folder).

Note

No prior knowledge of theoretical aerodynamics is required to use the application. However, a general understanding of aerodynamic quantities (lift coefficient C_L, drag coefficient C_D, lift-to-drag ratio C_L/C_D, Reynolds number, angle of attack α) makes interpreting the results easier.

1.3 Functional architecture

The application is structured around **three tabs** corresponding to as many logical steps:

1. Airfoils — loading, editing and comparison of geometric airfoils. Two airfoils are handled simultaneously: an editable *current airfoil* (in blue), and a *reference airfoil* (in red) that serves as a comparison.

2. XFoil Settings — entry of the simulation parameters: Reynolds range, angle-of-attack sweep, boundary layer parameters, panel options.

3. Results — graphical presentation of the results as a *configurable grid of analysis cells*, each displaying an independent plot (polar $C_L(\alpha)$, $C_D(\alpha)$, lift-to-drag ratio, pressure distribution $-C_p(x)$, boundary layer profile, etc.).

1.4 Airfoil representation

AirfoilTools offers two airfoil representation modes:

Discrete points mode

The airfoil is defined by a sequence of points in the **Selig** convention (running from the trailing edge to the upper surface, then the leading edge, then the lower surface, and back to the trailing edge). This is the most common format of airfoil libraries (for example `airfoiltools.com`).

Spline mode (multi-segment Bézier)

The airfoil is defined by two **BezierSpline** (upper surface and lower surface), each composed of several Bézier segments joined with C^1 continuity. This mode allows:

- *interactive editing* by moving the control points;
- *on-demand resampling* with a freely chosen point density;
- the *exact computation of the curvature and tangents* at any point;
- the *splitting of a segment* into two at an arbitrary point (De Casteljau algorithm, mathematically exact).

Switching from one mode to the other is done via the Edit → Convert to Spline menu or by loading a file in `.bspl` or `.bez` format.

Tip

To fully benefit from the interactive editing features and curvature plotting, it is recommended to **convert your airfoils to Spline mode** as soon as they are loaded. The following chapters detail this procedure.

2

Installation and launch

2.1 Prerequisites

- **Operating system :** Windows 10 or 11 (tested), Linux and macOS (compatibility expected, not tested by the developers).
- **Python :** version 3.10 or higher (recommendation: 3.13).
- **XFoil solver :** required only for running simulations. On Windows, the executable `xfoil.exe` is provided in the `externaltools/xfoil/` folder of the distribution.

2.2 Windows executable distribution

For Windows users who do not wish to install Python, a standalone `.exe` distribution is available. It bundles the Python interpreter, all dependencies and the XFoil binary.

The application is launched by double-clicking on `AirfoilTools.exe`.

> **Note**
>
> On first launch, Windows may display a *SmartScreen* warning because the executable is not signed. Click on “More info ” then “Run anyway ” to allow it to launch.

2.3 Installation from source

2.3.1 Retrieving the repository

```
git clone https://github.com/BenoitGFree/AirfoilTools.git
cd AirfoilTools
```

Listing 2.1: Cloning the repository

2.3.2 Creating the Python environment

```
py -3 -m venv env_py3
env_py3\Scripts\activate
pip install -r requirements.txt
```

Listing 2.2: Creating the venv and installing the dependencies (Windows)

The `requirements.txt` file lists the dependencies :

- `numpy` — vector computation ;
- `scipy` — numerical algorithms (interpolation, optimization) ;
- `matplotlib` — 2D plotting ;
- `PySide6` — Qt for Python graphical interface.

2.3.3 Launching the application

```
env_py3\Scripts\python.exe run_gui.py
```

Listing 2.3: Starting the GUI

The application opens on the main window shown in figure 1.1.

2.4 Verifying the installation

To verify that the environment is correctly configured, it is possible to run the unit test suite :

```
env_py3\Scripts\python.exe -c "import unittest; import sys; ^
sys.path.insert(0, 'sources'); sys.path.insert(0, 'tests'); ^
from test_bezier import *; ^
unittest.main(module='test_bezier', exit=False, verbosity=2)"
```

Listing 2.4: Running the tests

The full suite contains more than 240 tests, all of which must pass successfully.

Caution

Automatic detection of the XFoil executable depends on the location of `xfoil.exe`. The application looks for it in this order :

1. next to the application executable (PyInstaller *frozen* mode) ;
2. in the `externaltools/xfoil/` folder ;
3. in the system PATH.

If XFoil is not found, the XFoil Settings tab works but launching a simulation will fail.

3

General interface

3.1 Overview

The main window of AirfoilTools is organised into four areas (figure 1.1):

1. the **menu bar** (File, Edit, View, Options, Help);
2. the **tab selector** (Airfoils, XFoil Settings, Results, Cp / Boundary layer);
3. the **work area**, which changes depending on the selected tab;
4. the **status bar**, at the bottom of the window, where information messages and simulation progress are displayed.

3.2 Tooltips

Each control (checkbox, input field, button, menu action) has a **tooltip** that appears when the mouse cursor hovers over it for a moment. These tooltips describe the role of the control and the typical values.

Tip

Hover over a control whose function is not obvious: the tooltip appears after about a second. For menu actions, the help is displayed at the bottom of the window in the status bar when hovering over the menu entry.

3.3 File menu

Entry	Shortcut	Action
Open current airfoil...	Ctrl+O	Loads an airfoil into the current position (blue). Accepted formats: `.dat` (Selig/Lednicer), `.bspl` (AirfoilTools spline), `.bez` (AirfoilTools Bézier), `.csv`.
Open reference airfoil...	Ctrl+Shift+O	Loads an airfoil into the reference position (red).
From the UIUC database / Current airfoil...	—	Opens a search dialog in the online UIUC (Selig) database and downloads the selected airfoil. See section 4.3, page 15.
From the UIUC database / Reference airfoil...	—	Same for the reference airfoil.
NACA airfoil / Current airfoil...	—	Generates a NACA airfoil (4 or 5 digits) as the current airfoil. Prompts for the digits and then for the number of points. See section 4.4, page 17.
NACA airfoil / Reference airfoil...	—	Same for the reference airfoil.
Save airfoil...	Ctrl+S	Saves the current airfoil. A first dialog lets you save the *full airfoil*, the *upper surface* only or the *lower surface* only; the format is then selected from the filter of the file dialog.
Save airfoil with flap...	—	Saves the airfoil with the flap deflected (green), in the same way as the current airfoil. Requires the flap to be active. See section 4.7, page 20.
Open project...	Ctrl+Shift+P	Opens an AirfoilTools project (`.aftproj`): restores the current and reference airfoils as well as the tracing image and its transformation. See chapter 8.
Save project...	Ctrl+Alt+S	Saves the whole session (airfoils + tracing image) to a `.aftproj` file.
Load tracing image...	—	Loads an image (jpg, png, tif, bmp) in the back-

3.4 Edit menu

Entry	Shortcut	Action
Convert to Spline	Ctrl+B	Converts the current airfoil (discrete points mode) to multi-segment spline mode. See the section *Conversion to spline*, page 18.
Sampling / Current airfoil…	—	Changes the number of sampling points of the current airfoil. Available only in Spline mode.
Sampling / Reference airfoil…	—	Same for the reference airfoil.

3.5 View menu

Entry	Shortcut	Action
Fit zoom	Ctrl+0	Reframes the Airfoils canvas view on all visible airfoils.
Layout / 1×1, 1×2, …	—	Configures the grid of the Results tab (rows × columns).

3.6 Options menu

Entry	Shortcut	Action
Deviation / Vertical (thickness)	—	Computes the deviation between airfoils as a thickness difference measured **vertically** (along the z axis), at constant abscissa. Default mode.
Deviation / Normal (perpendicular)	—	Computes the deviation **perpendicularly** to the surface of the current airfoil (distance along the normal). See section 4.2.5, page 14.

Note

The two deviation modes are **exclusive**: choosing one automatically unchecks the other. The change is reflected immediately on the plot of the Airfoils tab.

3.7 Help menu

Entry	Shortcut	Action
User manual	F1	Opens this manual (PDF file) with the system's default viewer. The PDF is embedded in the `.exe` distribution and remains accessible even offline.
About…	—	Displays version information and the author.

Note

In development mode (launched via `run_gui.py`), the manual is read from `docs/manuel/manuel.pdf` in the project tree. In PyInstaller executable mode, it is extracted from the internal bundle (folder `_internal/docs/manuel.pdf` next to the exe).

3.8 Tabs

The four tabs are independent: switching tabs does not lose any data. The status bar can display contextual messages depending on the active tab. The Cp / Boundary layer tab (chapter 7) provides a detailed, *XFoil*-style view of a single operating point.

Stale results indicator

The Results tab has a special mechanism: when an airfoil is modified after the simulation has been run, a **yellow banner** appears at the top of the tab to indicate that the displayed results no longer match the current airfoils. It is then recommended to re-run the simulation.

Note

The staleness banner does not block consultation of the results: it is merely a visual indicator. The banner is removed automatically when a new simulation is launched.

4

Airfoils Tab

The Airfoils tab is the heart of the application: it allows loading, visualizing, editing and comparing airfoils. At startup, the current airfoil (blue) is a **NACA 2412** automatically **converted to Bézier**, and the reference airfoil (red) is a **NACA 2412** in discrete-points mode. By default, the **curvature** of the current airfoil, the **deviation** between the two airfoils and the **flap** (deflected at $-10°$) are also displayed.

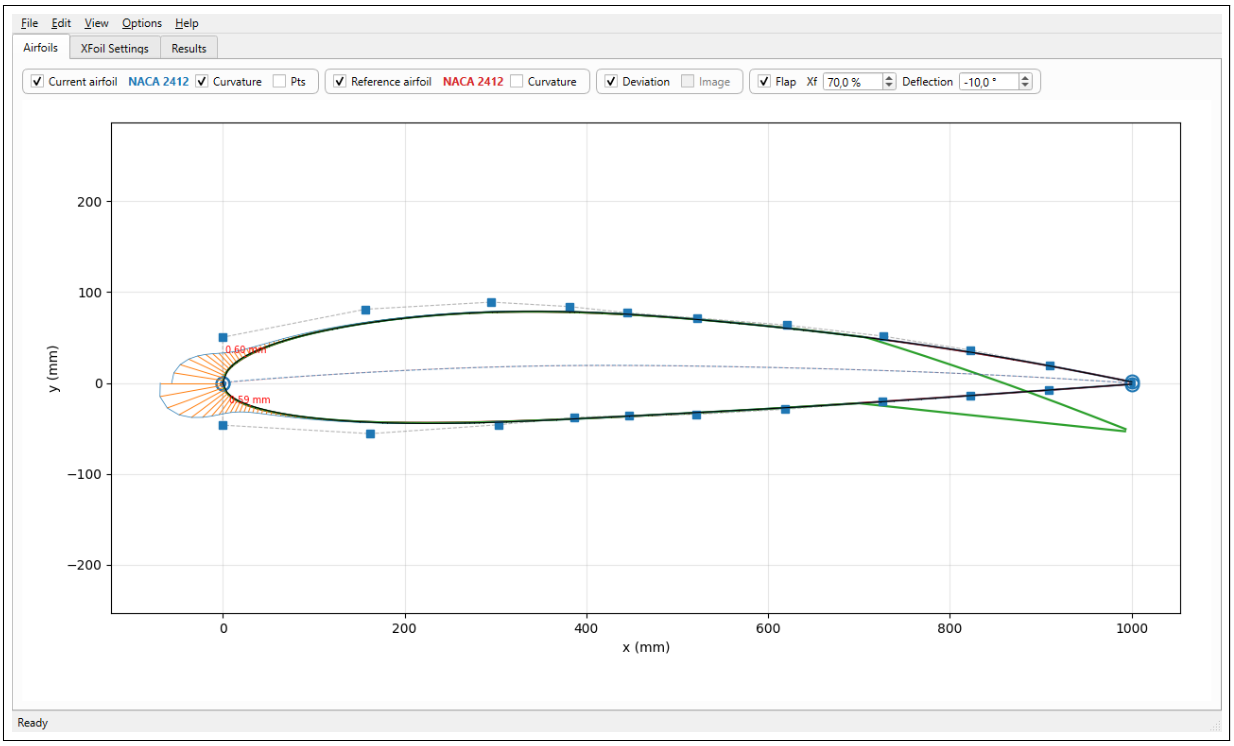

Figure 4.1: Airfoils tab in its default state: current airfoil (blue, Bézier) with its control points, curvature at the leading edge, airfoil with flap (green) and deviation relative to the reference.

4.1 Control bar

The bar located below the tab selector groups the display options, organized in **four frames** according to whether they concern a single airfoil, both, or none.

"Current airfoil " frame

- **Current airfoil** (check box): shows or hides the current airfoil. The colored label indicates its name (by default "NACA 2412 ").
- **Curvature**: displays the curvature *porcupines* of the current airfoil (segments perpendicular to the curve, length proportional to the local curvature).

- **Pts**: displays the points actually sampled on the splines (markers).

"Reference airfoil " frame

- **Reference airfoil** and **Curvature**: same for the reference airfoil.

Global frame (two airfoils / none)

- **Deviation**: plots the geometric differences between current airfoil and reference. The computation **mode** (vertical or normal) is chosen in the Options → Deviation menu. See section 4.2.5, page 14.
- **Image**: shows or hides the **tracing image** (background) when an image has been loaded. The box remains inactive (greyed out) as long as no image is loaded. See section 4.8, page 22.

"Flap " frame

- **Flap** (check box): activates the generation and display of an airfoil with deflected flap (green).
- **Xf**: position of the hinge axis, in percent of chord from the leading edge.
- **Deflection**: flap deflection angle, in degrees (positive = trailing edge upward).

The flap module is detailed in section 4.7, page 20.

> **Important**
>
> The **Curvature** options only work on airfoils in **spline mode**. If you try to activate them on an airfoil in discrete-points mode, a warning message is displayed and the box unchecks itself automatically (see figure 4.13).

4.2 Airfoil visualization

4.2.1 Discrete-points mode

This is the default mode at loading. The airfoil is plotted as a smooth curve interpolated between the discrete points. No control point is visible.

4.2.2 Spline mode

Once the airfoil is converted to a spline (see section 4.5), numerous elements appear:

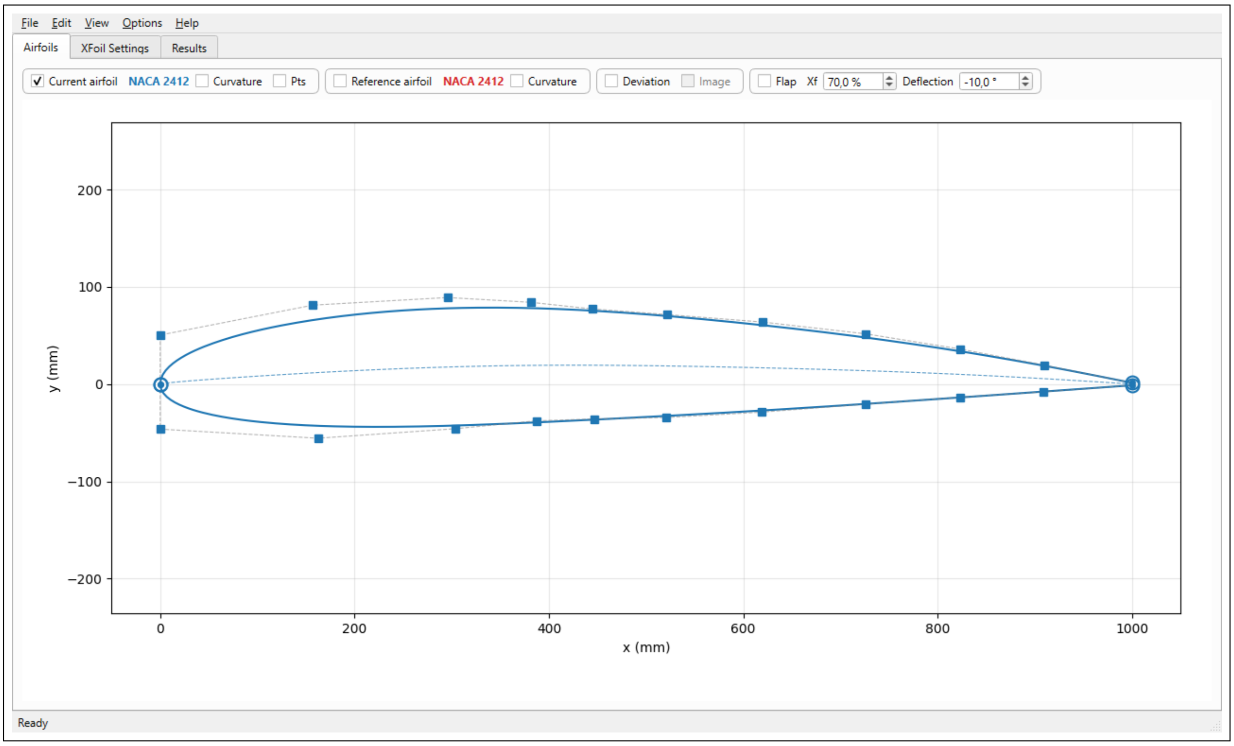

Figure 4.2: Airfoil in spline mode: the blue squares are the interior control points of the Béziers; the points surrounded by a circle mark the joins between consecutive segments.

- **Blue squares**: interior control points of the Bézier curves. They are **draggable** with the mouse to modify the airfoil shape.
- **Circles with a central point**: join points between two consecutive Bézier segments. They are also draggable; their movement preserves C^1 continuity with the neighboring segments.
- **Control polygon** (grey dotted lines): connects the control points in order.
- **Mean line** (light blue dashes): plots the relative camber of the airfoil (average of upper/lower surface).

4.2.3 Curvature display

Activating the **Curvature** box reveals the *porcupines*: segments perpendicular to the curve, whose length is proportional to the signed local curvature.

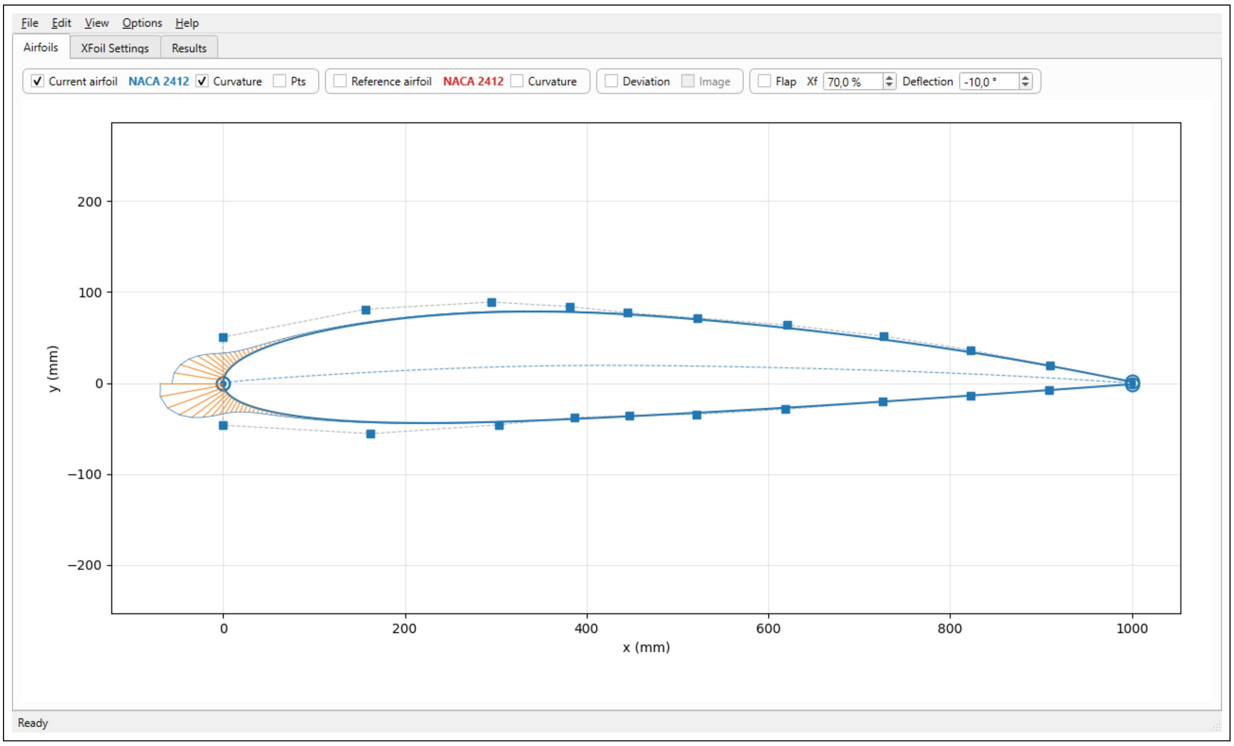

Figure 4.3: Display of the curvature porcupines (orange segments).

Note

Curvature is the inverse of the radius of the osculating circle. A positive curvature signals a curve concave toward increasing y (top of the upper surface), negative concave toward decreasing y. Curvature discontinuities are often visible at segment joins with C1 but not C2 continuity.

The **porcupine scale** and their **density** (number of quills per curve) are adjustable via the context menu (right-click on the canvas).

4.2.4 Display of sampled points

The **Pts** option reveals markers at the positions of the points actually used for discrete sampling. This is useful to check the density and distribution of the sampling before export or simulation.

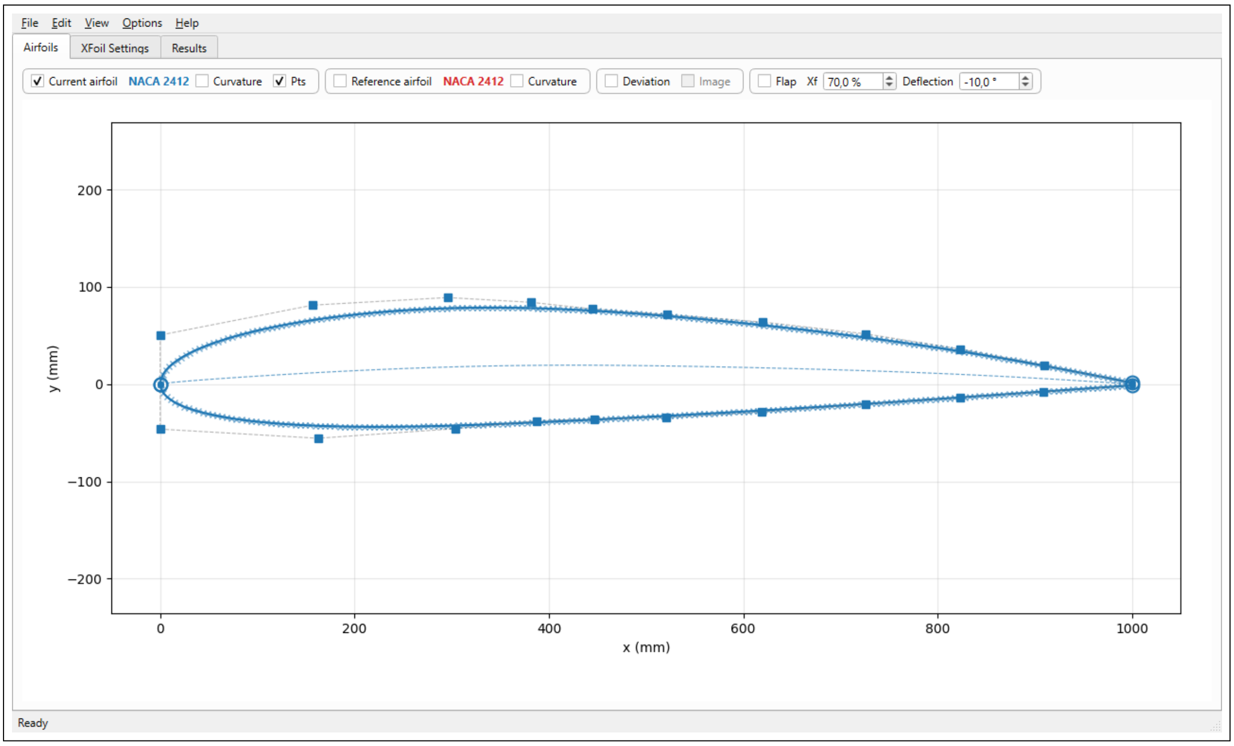

Figure 4.4: Display of the sampling points. The *curvilinear* mode (default) distributes the points uniformly along the arc.

4.2.5 Deviation visualization

The **Deviation** option measures the geometric difference between the current airfoil and the reference airfoil, as a bundle of segments (*quills*) accompanied by an envelope. The maximum deviation is highlighted.

Computation modes

The mode is chosen in the Options → Deviation menu:

- **Vertical (thickness)**: the difference is measured **vertically** (along the z axis), at constant abscissa x, by interpolation of the two airfoils on a common grid. This is the default mode, suited to thickness comparison.
- **Normal (perpendicular)**: the difference is measured **perpendicular to the surface** of the current airfoil (distance along the normal). This mode better reflects the actual geometric difference between two curves, in particular near the leading edge where the surfaces are strongly inclined.

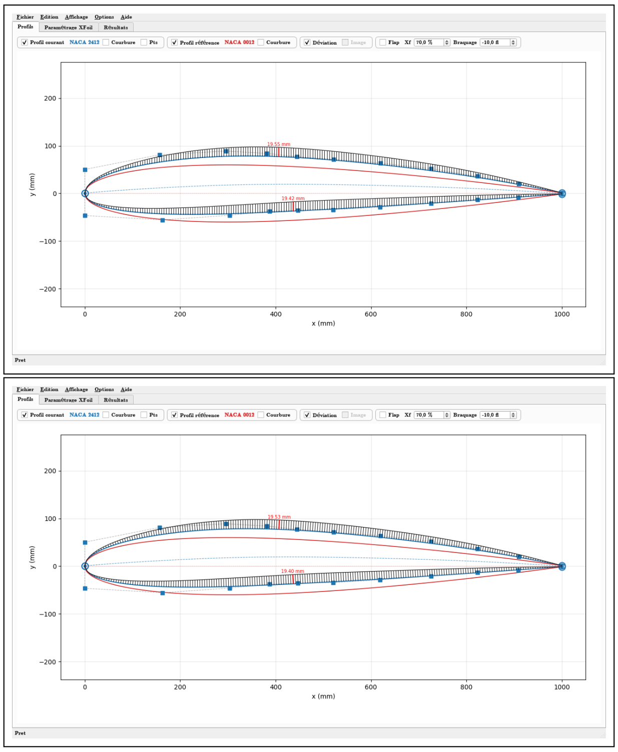

Figure 4.5: Deviation between NACA 2412 (current, blue) and NACA 0012 (reference, red). On the left the *vertical* mode (vertical quills), on the right the *normal* mode (quills perpendicular to the surface). In both cases, the maximum-deviation quill is plotted in red with its value in millimeters.

Maximum deviation

The **maximum** deviation is signaled by a **red** quill bearing its **value in millimeters** at its end. The computation is performed:

- **per Bézier segment** if the current airfoil is in spline mode (one maximum per upper-surface and lower-surface segment);
- **per side** (upper surface, lower surface) if the current airfoil is in discrete-points mode.

The displayed value is the **actual** deviation (not exaggerated), expressed in the unit of the airfoil (millimeters for a chord of 1 000 mm).

> **Tip**
>
> When the airfoils have a very small deviation to the eye (for example during an optimization by small displacements of the control points), it is useful to **amplify** the deviation display. The scale is adjustable via the context menu: Deviation scale.... This scale only affects the visual length of the quills, not the value in millimeters displayed at the maximum.

4.3 Loading an airfoil from the UIUC database

AirfoilTools provides **direct access** to the airfoil database of the University of Illinois (the **Selig** collection, about 1 600 airfoils), accessible via the entries File → From the UIUC database → Current airfoil... and ...Reference airfoil....

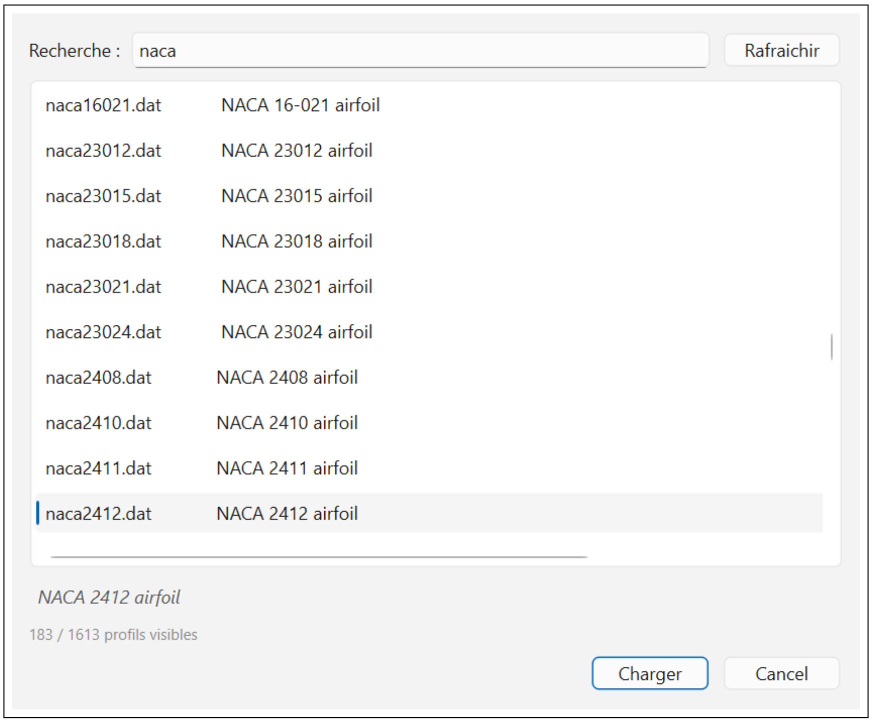

Figure 4.6: Loading dialog from the UIUC database. The search "naca " filters 183 airfoils out of 1 613. The selected airfoil shows its description below the list.

4.3.1 Usage

1. On the first call, the complete index (1 600 airfoils) is downloaded from the UIUC server. The download takes a few seconds depending on the connection.
2. Type all or part of the name or description of the airfoil in the Search field: the list filters in real time (case insensitive).
3. Click on an airfoil to see its description below the list. Double-click or the **[Load]** button to download and open it.
4. The airfoil is loaded in **discrete-points mode** (Selig format). To benefit from interactive editing, then use Edit → Convert to Spline (see next section).

4.3.2 Local cache

To avoid unnecessarily overloading the UIUC server and to allow offline use:

- The **index** (airfoil list + descriptions) is cached for **7 days**. Beyond that, it is re-downloaded automatically.
- The downloaded `.dat` **files** are cached **permanently** (UIUC airfoils are stable over time).

The **[Refresh]** button forces the re-download of the index to retrieve any new airfoils.
The cache location is:

- Windows: `C:\Users\<user>\.airfoiltools\cache\uiuc\`
- Linux / macOS: `~/.airfoiltools/cache/uiuc/`

Note

The `.dat` files of the UIUC database follow the Selig convention (path TE → upper surface → LE → lower surface → TE) with a header line containing the airfoil name. They are directly usable by AirfoilTools.

Caution

An internet connection is required for the first download of the index and of each airfoil. In case of unavailability of the UIUC server, the dialog displays an error message; the airfoils already cached, however, remain usable.

4.4 Generating a NACA airfoil

AirfoilTools can generate on the fly the airfoils of the **NACA** family from their designation, without going through a file. Generation is accessible via File → NACA airfoil → Current airfoil... (or ... Reference airfoil...).

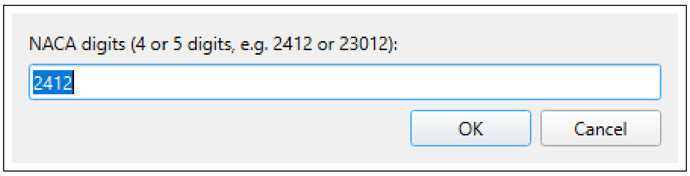

Figure 4.7: Entry of the NACA digits. A second dialog then asks for the number of points of the airfoil.

4.4.1 Usage

1. Enter the **digits** in the first dialog:
   - **4 digits** (e.g. 2412): maximum camber (%), position of max camber (tenths of chord), relative thickness (%);
   - **5 digits** (e.g. 23012): cambered airfoil with a more complex mean line.
2. Enter the **number of points** in the second dialog (default 150).
3. The airfoil is generated, normalized (chord 1 000 mm) and assigned to the chosen role (current or reference).

4.4.2 Clean regeneration from the context menu

An airfoil generated by this function is **marked** as a NACA airfoil as long as it has not been modified (editing of a control point) nor converted to a spline. For such a NACA airfoil in discrete-points mode, the **context menu** (right-click) offers the entry Number of points....

Changing the number of points this way does not merely interpolate the existing points: it **recalls the NACA generation function** to produce a *clean* airfoil at the new resolution, conforming to the analytical formula.

Note

This clean regeneration is only possible for an **unmodified** NACA airfoil. As soon as it is converted to a spline or edited, the marker is lost: one then reverts to the classic re-sampling of the spline.

4.5 Conversion to spline

To benefit from interactive editing, the airfoil must be converted to spline mode. This is done via:

- the Edit → Convert to Spline menu;
- the keyboard shortcut Ctrl + B.

A dialog opens to enter the conversion parameters:

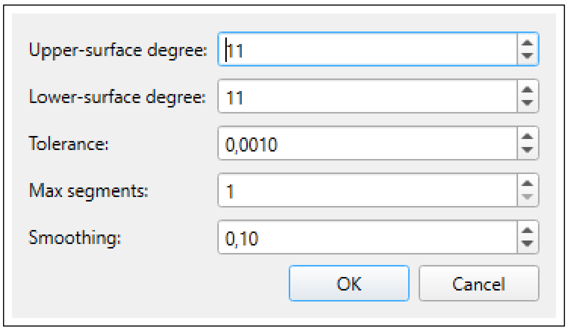

Figure 4.8: Conversion-to-spline dialog.

Parameter	Description
Upper-surface degree	Degree of the Bézier curves of the upper surface. Small (3–5) = very smoothed shape; medium (6–12) = compromise (recommended); large (> 15) = very precise shape but risk of oscillations.
Lower-surface degree	Same for the lower surface. Can differ from the upper-surface degree (typically the lower surface is less curved).
Tolerance	Maximum deviation accepted between the fitted spline and the discrete points, as a fraction of the chord. Typical value: 10^{-3} to 10^{-4}.
Max segments	Maximum number of Bézier segments per side. With 1, a single global segment is obtained (historical behavior). With a higher value and a fine tolerance, the *adaptive* algorithm splits at the point of greatest error until the tolerance is met, concentrating the segments in zones of high curvature (typically the leading edge).
Smoothing	Regularization coefficient of the fit (Tikhonov). 0 = pure least-squares fit; > 0 = smoothing of the control polygon.

Tip

For most airfoils, parameters of **degree 11**, **tolerance** 10^{-3}, **max segments 4**, **smoothing 0,1** give a good compromise between precision and geometric quality. Reduce the tolerance to 10^{-4} and increase *Max segments* to 8–10 for a very faithful approximation.

4.6 Interactive editing

In spline mode, the canvas becomes an **interactive geometric editor**.

4.6.1 Mouse interactions

Action	Effect
Left click + drag on a control point	Moves the control point. The curve is updated in real time.
Left click + drag in empty space	Draws a *zoom rectangle*. The view is reframed on the zone at the end of the drag.
Mouse wheel	Zoom centered on the cursor position.
Middle click (or Shift+left click)	Pan (translation of the view).
Right click	Opens the **context menu**.

4.6.2 Context menu (right-click)

The context menu offers several actions depending on the context:

- Point info... (if clicking on a control point): displays the coordinates and local parameters of the point.
- Release the vertical tangency constraint at the leading edge (if clicking on the P1 node, neighbor of the leading edge): releases or restores the vertical tangent at the leading edge. See section 4.6.3.
- Porcupine scale...: sets the display scale of the curvature.
- Porcupine density / ×2 or ÷2: modifies the number of quills displayed.
- Deviation scale... and Deviation density: same for the deviation porcupines.
- Split upper surface / Split lower surface (in spline mode): splits the spline in two at the nearest clicked point (De Casteljau algorithm, exact).
- Global parameters: number of sampling points, method (curvilinear or adaptive).
- Segment (if clicking within a specific segment): allows defining *per-segment overrides* (degree, number of points, method different from the global one).

4.6.3 Vertical tangent at the leading edge

By default, the first interior control point of each side (noted **P1**, neighbor of the leading edge) is **constrained** to remain on the vertical of the leading edge (LE). This constraint guarantees a **vertical tangent** at the leading edge, which corresponds to the natural geometry of most airfoils. During editing, the movement of P1 is therefore limited to the vertical axis.

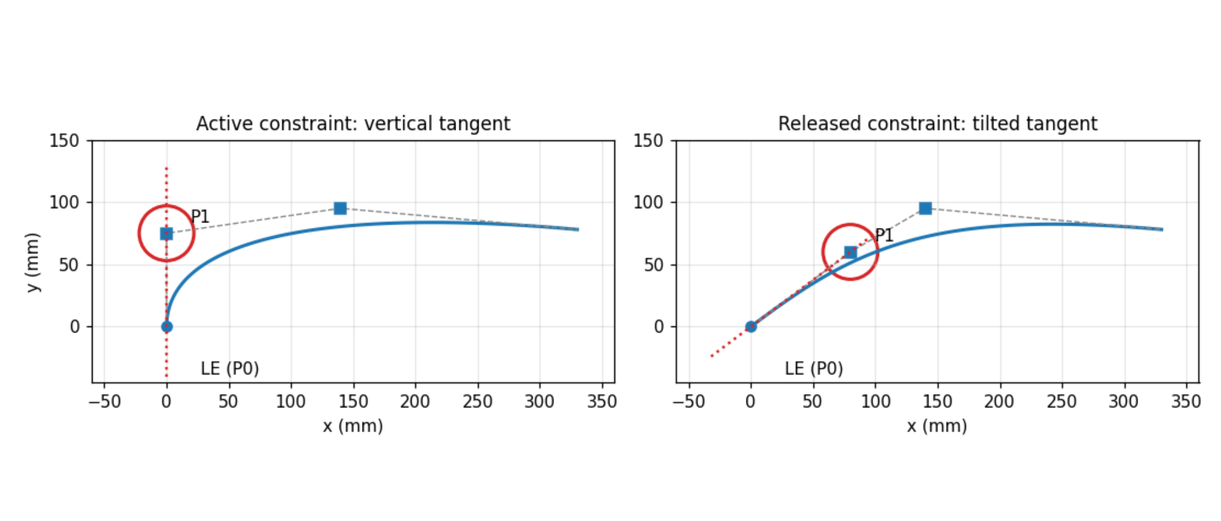

Figure 4.9: Effect of the vertical-tangent constraint at the leading edge. On the left, constraint active: the P1 node (circled in red) stays aligned vertically with the LE, the tangent (red dashes) is vertical. On the right, constraint released: P1 can be moved freely, the tangent tilts.

It is sometimes desirable to **release** this constraint (for example to trace an airfoil whose leading edge does not have a strictly vertical tangent). To do so:

1. Perform a **right-click** on the P1 node of the upper surface or lower surface.
2. Choose Release the vertical tangency constraint at the leading edge.

The P1 node then becomes freely draggable (in x and y). To restore the constraint, right-click again on the node and choose Restore the vertical tangency at the leading edge: the P1 node is automatically brought back onto the vertical of the leading edge.

Note

The constraint is managed **independently** for the upper surface and the lower surface. Releasing the tangent on the upper-surface side does not affect the lower surface, and vice versa.

4.7 Flap

The **flap** module generates a **deflected airfoil** from the current airfoil, by rotating its rear part around a hinge axis. The airfoil with flap is plotted in **green**, in addition to the current and reference airfoils; it is **not editable** (recomputed automatically) and can be simulated then saved like the current airfoil.

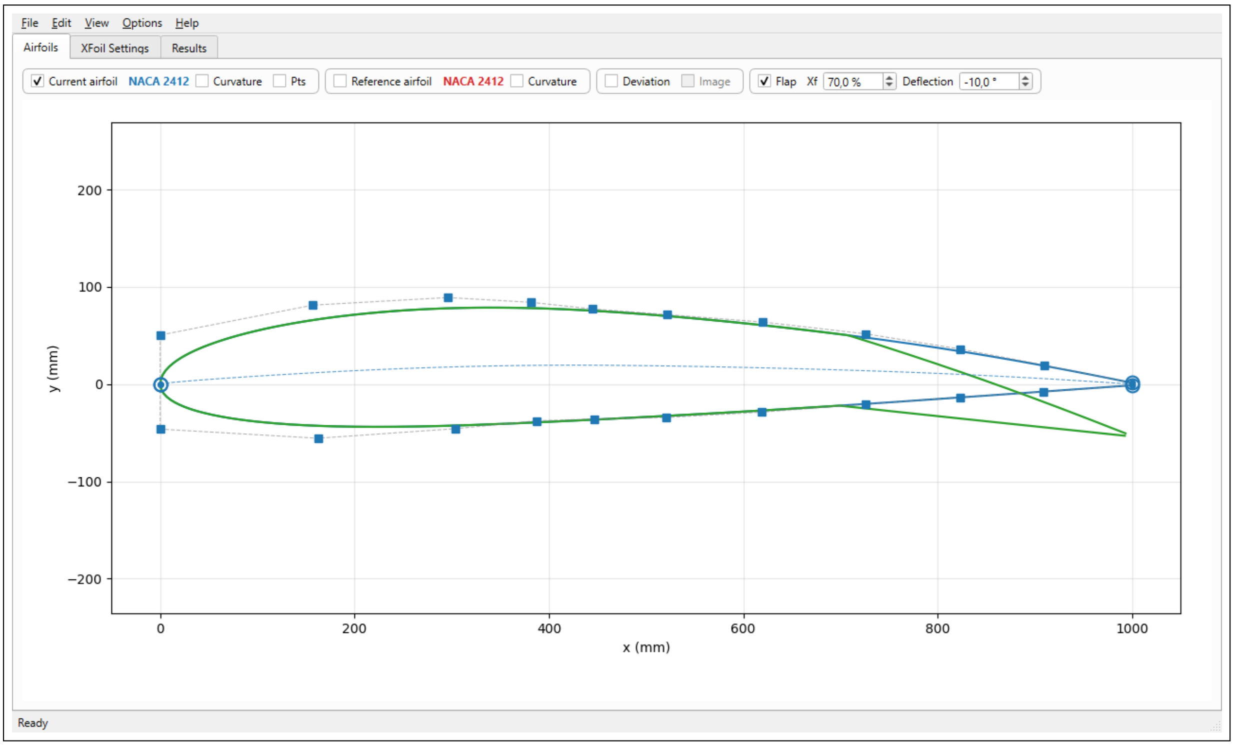

Figure 4.10: Airfoil with flap (green) deflected at $-10°$ at 70 % of chord, superimposed on the current airfoil (blue) and its control points.

4.7.1 Activation and settings

The controls are in the "Flap " frame of the Airfoils tab bar:

Control	Role
Flap	Activates the generation of the deflected airfoil (green).
Xf	Position of the hinge axis, in **percent of chord** measured from the leading edge (5 to 95 %).
Deflection	Deflection angle in degrees (-45 to $+45$). **Positive** = trailing edge **upward**; **negative** = trailing edge **downward**.

The deflected airfoil is recomputed in real time at each change of Xf or Deflection, and at each modification of the current airfoil.

4.7.2 Hinge geometry

The hinge axis is the **center of the inscribed circle** tangent simultaneously to the upper surface and the lower surface, whose abscissa equals Xf. The rear part of each surface is then rotated by the deflection angle around this axis.

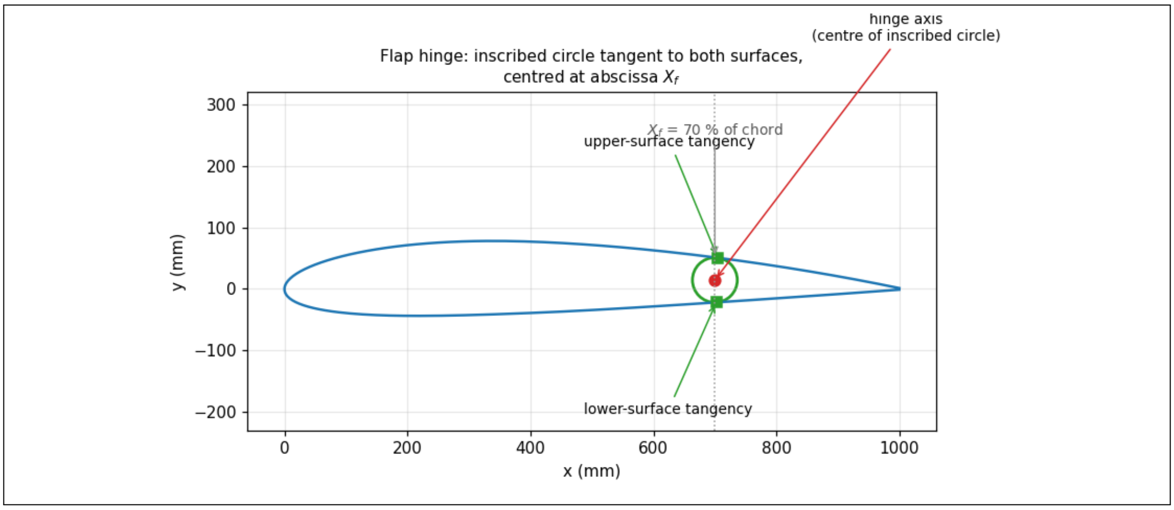

Figure 4.11: Construction of the hinge: the inscribed circle (green), tangent to the upper surface and the lower surface at the abscissa X_f, provides the rotation center (red point) of the flap.

Depending on the side and the sign of the deflection, the join between the fixed part and the deflected part is handled differently:

- on the side where the deflection **opens** a gap, an **arc of circle** fills the space;
- on the side where the deflection **closes back** the material, the surface is **cut at the intersection** (overlap removed).

4.7.3 Simulation and saving

- **Simulation**: the airfoil with flap is simulated by XFoil in the same way as the current and reference airfoils. The **[Flap only]** button of the XFoil Settings tab computes it in isolation; **[Run all]** includes it if it is active. The results appear under the *Flap* role (green) in the Results tab.
- **Saving**: File → Save airfoil with flap... saves the deflected airfoil, with the same complete / upper-surface / lower-surface choice and the same formats as the current airfoil.

Important

For the simulation, the airfoil with flap is expressed in the **normalized frame of the current airfoil**: XFoil neither re-normalizes nor re-aligns it. One thus observes only the effect of the deflection relative to the current airfoil (shift of the polar, of the moment, etc.), without any artifact of change of chord or of reference angle of attack.

4.8 Tracing image (tracing an airfoil)

AirfoilTools allows loading a **background image** (photo, scan of a drawing, screenshot of an airfoil) in order to **trace** its shape by moving the control points of the current airfoil until they match the silhouette of the image.

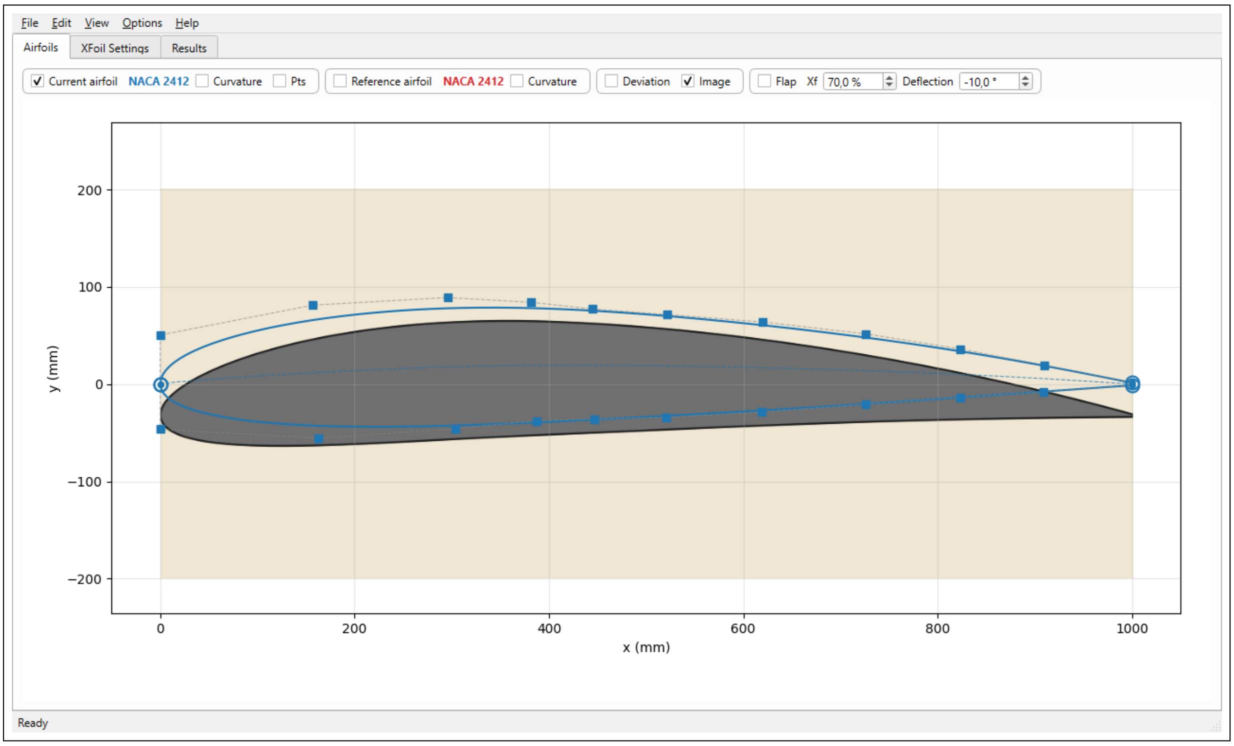

Figure 4.12: Tracing an airfoil: the tracing image (grey silhouette) is displayed in the background; the current airfoil (blue, in spline mode) and its control points are superimposed on top. The control points are adjusted to make the blue curve coincide with the silhouette.

4.8.1 Loading the image

Loading is done via File → Load tracing image.... The usual formats are accepted: `.jpg`, `.png`, `.tif`, `.bmp`. On opening, the image is scaled to cover approximately the chord (1 000 mm) and the **Image** box of the control bar becomes active and checked.

The image is drawn **in the background**, beneath the airfoil and its control points, which therefore remain perfectly visible and manipulable on top.

4.8.2 Positioning, scaling and rotating the image

The manipulations of the image are done by holding the i key down, combined with the mouse:

Action	Effect
i + left click dragged	**Moves** the image (translation).
i + mouse wheel	**Scales** the image (enlarges / reduces around its center).
i + right click dragged	**Rotates** the image around the origin $(0, 0)$ (the leading edge).
Shift + i + action	**Fine adjustments**: movement and rotation ten times slower, reduced scaling step. Ideal for precise alignment.

Tip

Proceed in two steps: first a coarse alignment (movement, scaling, rotation without Shift) to superimpose the image and the airfoil approximately, then a **fine adjustment** holding Shift for the precise alignment of the leading edge and the trailing edge.

Note

The canvas must have the **keyboard focus** for the i key to be taken into account: it obtains it automatically as soon as the cursor hovers over the plotting area.

4.8.3 Hiding or removing the image

- The **Image** box of the control bar shows or hides the image without deleting it.
- File → Remove the tracing image permanently removes the image from the session.

Important

The tracing image and its position (scale, rotation, translation) are **saved with the project** in the `.aftproj` format (see chapter 8). One thus finds the tracing exactly as it was upon reopening the project. The classic airfoil formats (`.dat`, `.bspl`. . .) do **not** contain the image: to keep the tracing, a project must be saved.

4.9 "Curvature unavailable " warning

When the Curvature box is activated while the concerned airfoil is in discrete-points mode (not converted to spline), an information message is displayed:

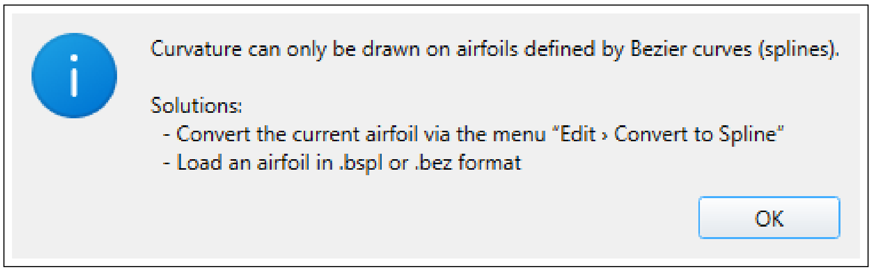

Figure 4.13: "Curvature unavailable " warning message.

As indicated in the message, two solutions are possible:

1. convert the airfoil to a spline (menu Edit → Convert to Spline);
2. load an airfoil in a format preserving the spline geometry (`.bspl` or `.bez`).

The box unchecks itself automatically after the message is displayed.

4.10 Modifying the number of points

In spline mode, the number of discrete sampling points of the splines can be modified via the Edit → Sampling → Current airfoil. . . menu (or . . . Reference airfoil. . .).

A dialog asks for the new number of points (between 10 and 10 000). This affects only the *export discretization* and the display: the underlying spline geometry is not altered.

Note

For XFoil simulations, the default sampling (150 points) is generally sufficient. XFoil then re-samples internally via the `PANE` command if the *Repanel* option is activated in the XFoil

> Settings tab.

5

XFoil Settings Tab

The XFoil Settings tab gathers all the simulation parameters passed to the solver. It is organized into input areas (Reynolds, Alpha, XFoil Parameters), a row of run buttons and a Diagnostics frame.

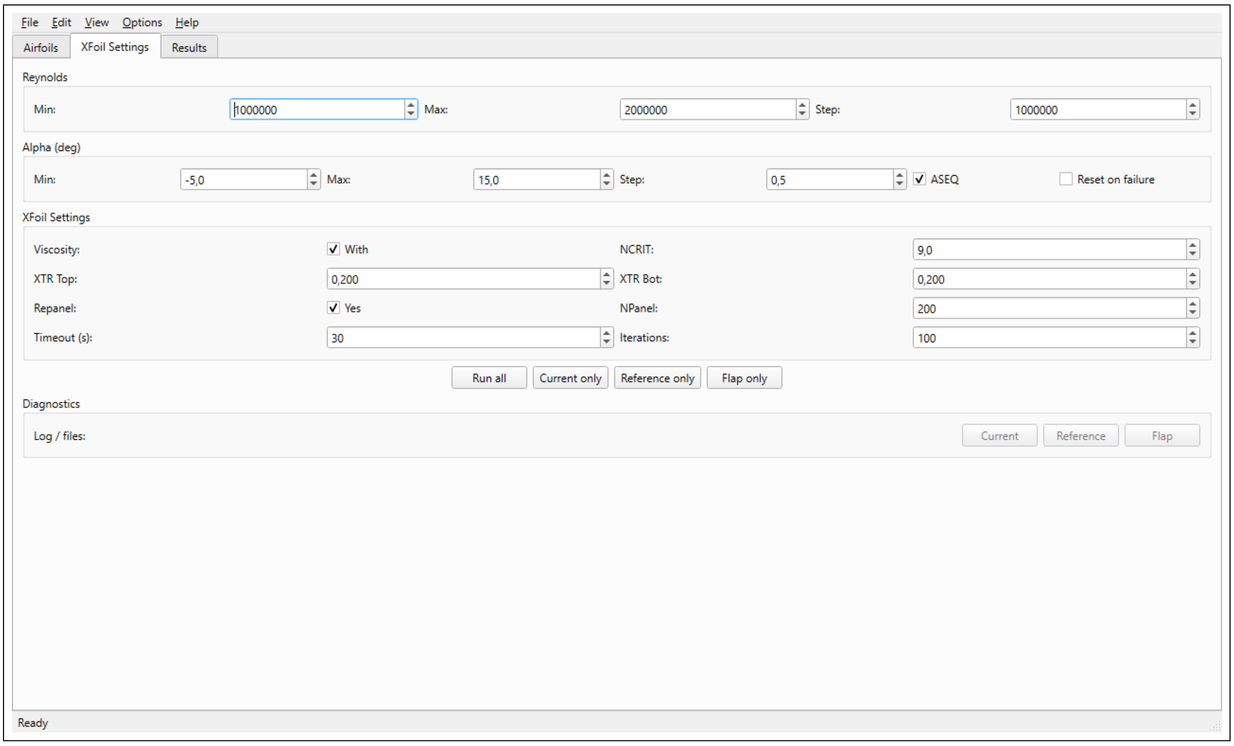

Figure 5.1: XFoil Settings tab: input areas, run buttons (including “Flap only ”) and “Diagnostics ” frame.

Note

Default values at startup: Reynolds from 1 000 000 to 2 000 000 (step 1 000 000), angle of attack from $-5°$ to $+15°$ (step 0,5°), forced transition XTR $= 0{,}2$, 100 iterations, ASEQ sweep.

5.1 Reynolds range

The Reynolds area lets you define a **range** of Reynolds numbers for the sweep. For each Reynolds, a complete polar is computed. Three fields:

Field	Description
Min	Minimum Reynolds of the sweep.
Max	Maximum Reynolds of the sweep.
Step	Increment between two consecutive values. To compute a single Reynolds only, choose `Step` $\geq$ `Max - Min`.

5.2 Angle of attack range (Alpha)

The Alpha (deg) area defines the sweep in angle of attack α for each polar:

Field	Description
Min	Minimum angle of attack (in degrees, may be negative).
Max	Maximum angle of attack.
Step	Increment between two points of the polar. Smaller = finer polar, but longer simulation.

ASEQ mode

The ASEQ checkbox controls the **sweep strategy** in angle of attack:

- **Checked** (default): XFoil runs the `ASEQ` command which automatically sweeps the range starting from the minimum. This is the historical *fast* mode.
- **Unchecked**: AirfoilTools sends a succession of individual `ALFA` commands, starting from $\alpha = 0$ towards the maximum, then from $-\Delta$ towards the minimum (with an `INIT` command between the two). This mode is more *robust* for convergence at high angles or for difficult airfoils.

Tip

If XFoil does not converge for certain high angles, disable ASEQ to try the bidirectional sweep. This often improves convergence by initializing each branch from the solution at $\alpha = 0$.

Reset on failure option

When an angle-of-attack point does not converge (message `VISCAL: Convergence failed`), the running sweep may **abandon the entire remainder** of the branch (cascade effect). The Reset on failure checkbox adds a **second pass** that replays each angle of attack from a clean state (`INIT` command before each `ALFA`), in order to recover the missed points without degrading the running sweep (the postprocessor keeps the solution obtained while running when it exists).

Caution

The Reset on failure option implies the individual `ALFA` mode (incompatible with ASEQ) and **roughly doubles** the computation time. Remember to increase the Timeout accordingly. It is **unchecked by default**.

5.3 Advanced XFoil parameters

The XFoil Parameters area gathers the options of the solver itself.

5.3.1 Viscosity

- Checked (With): **viscous** analysis (XFoil's `VISC` command), with boundary layer. **Recommended for any realistic polar.**
- Unchecked: inviscid analysis (potential, without friction). Useful only for quick geometric comparisons.

5.3.2 NCRIT

Boundary layer transition criterion by the e^N method (amplification of Tollmien-Schlichting waves).

NCRIT	Flow type
9	Standard free flow (default value).
4–7	Very turbulent flow (closed wind tunnels, high ambient turbulence).
12–14	Very calm flow (glider flight in perfectly laminar air).

5.3.3 XTR Top and XTR Bot

Position of **forced transition** of the boundary layer, respectively on the upper surface (*top*) and lower surface (*bottom*), expressed as a fraction of chord (x/c).

- Value 1,0: free transition, computed by the NCRIT criterion. This is the standard mode.
- Value < 1,0: forced transition at the indicated position (simulates a turbulent *trip*: turbulator, roughness, insect impacts. . .).

Important

The minimum value is **0,05**. Forcing the transition too close to the leading edge (for example 0,01) **destabilizes the XFoil boundary layer**: the transition routine diverges (`NaN`) and enters a near-infinite loop, until the Timeout expires. The default value is **0,2**.

5.3.4 Repanel and NPanel

- Repanel: if checked, XFoil runs the `PANE` command which re-samples the airfoil with a curvature-based adaptive criterion. Recommended for airfoils with non-uniform sampling or for very high resolution airfoils.
- NPanel: number of panels after re-sampling (default 200). More = more accurate but slower.

5.3.5 Timeout

Maximum execution time of XFoil for the computation of one polar, in seconds (default 30 s). Beyond that, the process is killed. This prevents the application from staying blocked if XFoil enters a non-convergence loop.

Caution

Increase the timeout (for example to 120 s) if you sweep a very wide range of angles of attack ($\Delta\alpha > 30°$), if you compute several Reynolds, if the Reset on failure option is active (doubled time), or if your airfoils converge slowly.

5.3.6 Iterations

Maximum number of Newton iterations per point (XFoil's `ITER` command), default 100. **Increase** this value (200–300) when a specific point does not converge (`VISCAL: Convergence failed`), a typical situation near stall or with a flap hinge.

5.4 Run buttons

Four buttons at the bottom of the parameters area let you run the simulation:

- **[Run all]**: simulates all active airfoils — current, reference and, if the flap is enabled, the airfoil with flap.
- **[Current only]**: simulates only the current airfoil (blue).
- **[Reference only]**: simulates only the reference airfoil (red).
- **[Flap only]**: simulates only the airfoil with flap (green). Requires the flap to be enabled in the Profils tab.

During the simulation, the tab controls are disabled, and the status bar shows the progress. At the end, the Results tab is selected automatically.

Important

If you re-run a simulation for **only one** of the airfoils, the previous results of the other airfoils are *kept* (merge). This allows, for example, fast iteration on the current airfoil while keeping the reference fixed for comparison.

5.5 Diagnostics

The Diagnostics frame, at the bottom of the tab, gives access to the internal files of the last simulation of each airfoil (**[Current]**, **[Reference]**, **[Flap]**). The buttons remain inactive as long as no simulation has been run for the corresponding airfoil.

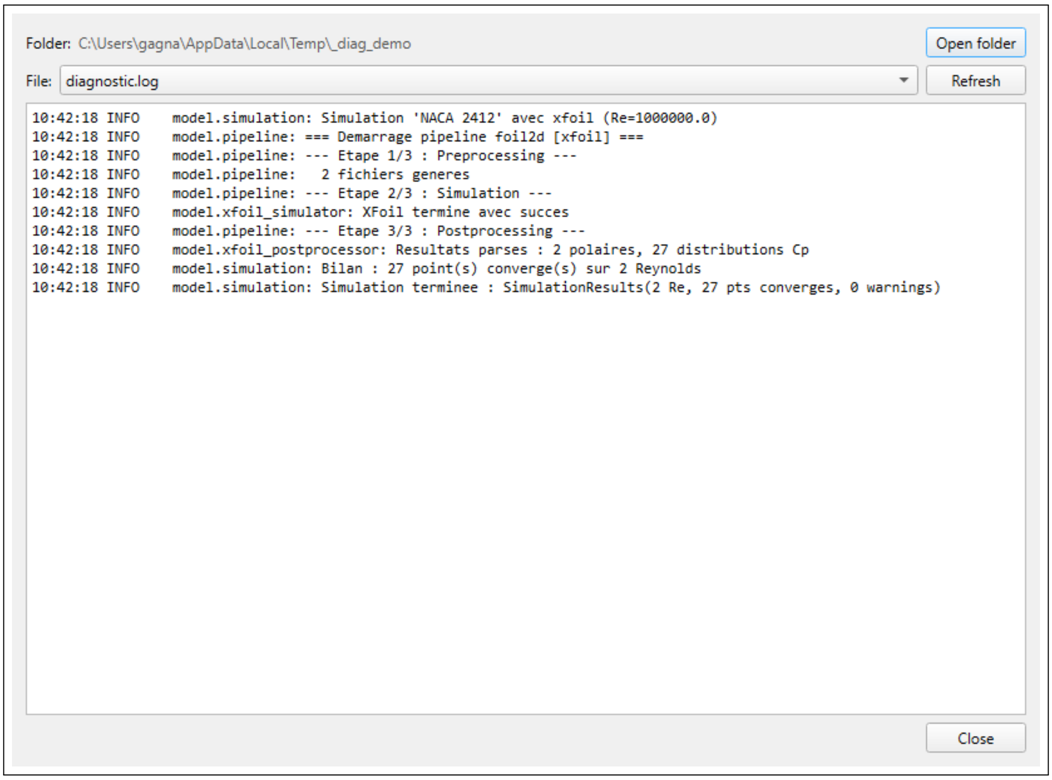

Figure 5.2: Diagnostics window: access to the working directory and the simulation files. The `diagnostic.log` log summarizes the course of the computation.

The diagnostics window lets you:

- browse the simulation **files**: the `diagnostic.log` log (summary of the course: steps, warnings, convergence report), the XFoil console log (`xfoil_alpha.cmd.log`), the command script, polars, pressure distributions;
- **open the working folder** in the system file explorer.

Tip

In case of failure or incomplete results, first open `diagnostic.log`: it clearly indicates whether the problem comes from a non-convergence ("NO point converged. . . "), a *timeout*, or an empty polar. The XFoil console log then provides the detail angle of attack by angle of attack. The log is saved **even in case of timeout**.

6

Results Tab

The Results tab presents simulation results as a **configurable grid of analysis cells**. Each cell displays an independent plot type (polar, pressure distribution, boundary layer profile, etc.).

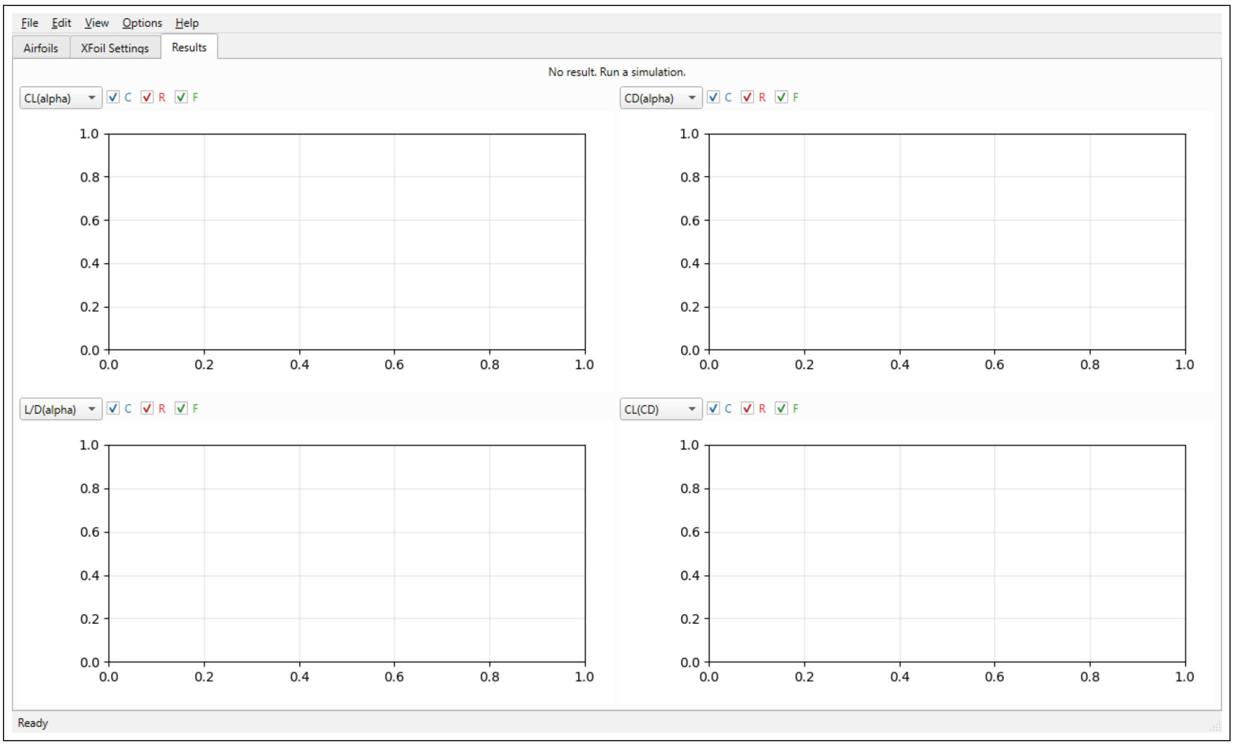

Figure 6.1: Results tab before any simulation. The grid defaults to a 2×2 layout, i.e. four empty cells.

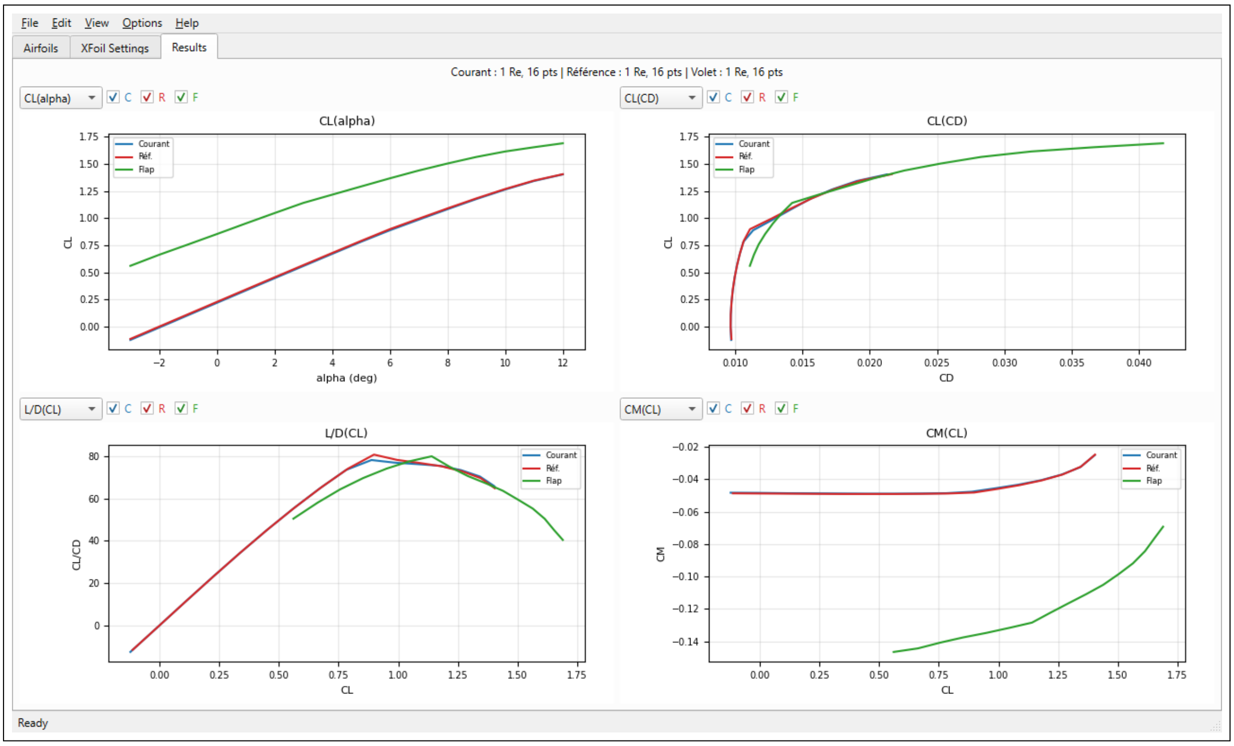

Figure 6.2: Results tab after simulating the current airfoil (blue), the reference (red) and the airfoil with flap (green). The four cells here display $C_L(\alpha)$, the $C_L(C_D)$ polar, the lift-to-drag ratio C_L/C_D as a function of C_L and the $C_M(C_L)$ moment.

6.1 Grid configuration

The grid is configurable via the View → Layout menu. Five preconfigured layouts are available: 1×1, 1×2, 2×2 (default), 2×3, 3×3.

Note

When the layout is changed, the analyses already selected in existing cells are kept. New cells are initialized with the default analyses.

6.2 Analysis cell

Each cell features a **control bar** and a **matplotlib canvas**.

6.2.1 Control bar

Element	Role
Drop-down list (analysis)	Choice of the analysis type to display in the cell (see *Analysis types*, below).
C checkbox	Shows / hides the current airfoil curves (blue).
R checkbox	Shows / hides the reference airfoil curves (red).
F checkbox	Shows / hides the airfoil with flap curves (green).
Reynolds list	For analyses that depend on a particular Reynolds ($-C_p(x)$, Cp + Airfoil, airfoil + CL), selects the Reynolds to display. Hidden if not relevant.
Alpha list	For analyses that depend on an angle of attack (Cp + Airfoil, airfoil + CL and the boundary layer quantities), selects the angle of attack α to display. Hidden if not relevant.
Coordinates	Real-time display of the cursor coordinates in the plot frame.

6.2.2 Mouse interactions

The interactions available on the matplotlib canvas are identical to those of the Airfoils tab:

- Left click + drag: rectangle zoom;
- Wheel: zoom centered on the cursor;
- Middle click: pan;
- Left click on the legend: hides / shows the corresponding curve.

6.2.3 Interactive and scrollable legend

When a cell contains many curves (for example several Reynolds combined with several airfoils), the legend can become too long to be displayed entirely. A **scroll bar** appears automatically beyond 15 entries.

Tip

To quickly compare two curves, it is more efficient to **hide** the interfering curves by clicking on their entry in the legend, rather than zooming or adding an extra cell.

6.3 Available analysis types

6.3.1 Polars (as a function of Alpha)

Analysis	Description
$C_L(\alpha)$	Lift coefficient as a function of the angle of attack. One curve per Reynolds.
$C_D(\alpha)$	Drag coefficient as a function of the angle of attack.
$C_M(\alpha)$	Pitching moment coefficient (quarter-chord reference).
$C_L/C_D\ (\alpha)$	*Lift-to-drag ratio* as a function of the angle of attack. Allows the angle of attack of maximum lift-to-drag ratio to be determined.
$C_L/C_D\ (C_L)$	*Lift-to-drag ratio* as a function of the lift coefficient. Convenient for reading the lift-to-drag ratio at the targeted cruise C_L.
$C_M(C_L)$	Moment coefficient as a function of the lift. The slope indicates stability; the effect of a flap is very readable here.
$C_D(C_L)$ *(Lilienthal polar)*	Classic polar: drag on the x-axis vs lift on the y-axis. The tangent from the origin gives the maximum lift-to-drag ratio.
Transition top, bot	Position x/c of the boundary layer transition on the upper / lower surface, as a function of α. Indicates the effective roughness and laminar separation.

6.3.2 Distributions along the airfoil

Analysis	Description
$-C_p(x)$	Distribution of the pressure coefficient along the chord, at a given Reynolds and angle of attack. Convention: $-C_p$ upward (upper surface on top).
Cp + Airfoil	Combined *XFoil*-style plot: $-C_p$ on top and the airfoil redrawn below, for a (Reynolds, angle of attack) pair. An inset summarizes the coefficients (C_L, C_M, C_D, lift-to-drag ratio). For a fuller dedicated view (inviscid C_p and boundary layer), see chapter 7.
Airfoil + CL	Combined visualization: airfoil rotated by the angle of attack + plot of the boundary layer edge (displacement thickness δ^*).

6.3.3 Boundary layer

Five boundary layer quantities can be plotted as a function of the curvilinear abscissa s or the Cartesian abscissa x. They are evaluated at the **selected angle of attack** (Alpha list) — the boundary layer being computed for each α — and each curve compares the available Reynolds numbers:

Quantity	Description
$U_e(s)$, $U_e(x)$	Velocity at the edge of the boundary layer, normalized by the upstream free-stream velocity.
$\delta^*(s)$, $\delta^*(x)$	Displacement thickness.
$\theta(s)$, $\theta(x)$	Momentum thickness.
$C_f(s)$, $C_f(x)$	Local skin friction coefficient.
$H(s)$, $H(x)$	Shape factor $H = \delta^*/\theta$ ($H \approx 2{,}6$ at the transition point, $H \approx 4$ at separation).

6.4 Stale results banner

When the user returns to the Airfoils tab and modifies an airfoil after simulation (moving a control point, changing the degree, conversion…), a yellow banner appears at the top of the Results tab:

> △ *The current airfoil has been modified since the last simulation. Re-run to update.*

This banner disappears automatically at the next simulation.

> **Important**
>
> The results displayed in the cells are *not* recomputed automatically after an airfoil is modified. The banner is a simple visual indicator: it is up to the user to re-run the simulation from the XFoil Settings tab.

7

Cp / Boundary Layer Tab

The Cp / Boundary layer tab provides a detailed, *XFoil*-style view of a **specific operating point**: one airfoil, one Reynolds number and one angle of attack. It gathers in a single window the pressure distribution (viscous *and* inviscid) and the airfoil geometry surrounded by its boundary layer.

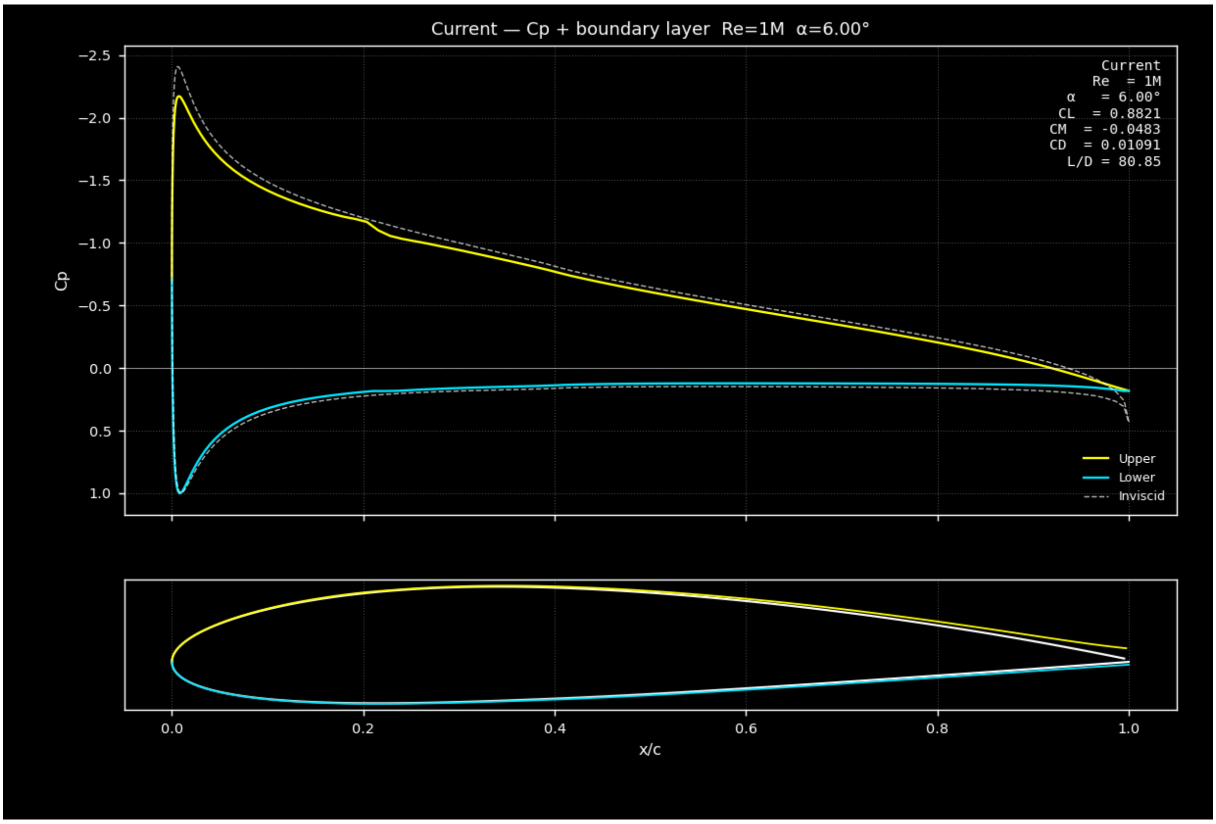

Figure 7.1: Cp / Boundary layer tab for a NACA 2412 at $Re = 10^6$ and $\alpha = 6$ř. Top: the pressure coefficient C_p (upper surface in yellow, lower surface in cyan) and its inviscid value (white dashed line). Bottom: the airfoil to scale, with the boundary layer edges δ^* on the upper surface (yellow) and lower surface (cyan). The inset summarizes the coefficients at the selected point.

7.1 Selecting the operating point

A control bar at the top of the tab targets the calculation to display:

List	Role
Airfoil	Airfoil to display: current (blue), reference (red) or flap (green). Only the airfoils actually simulated are offered.
Re	Reynolds number of the calculation.
α	Angle of attack (in degrees) of the calculation.

The lists are **cascading**: changing the airfoil updates the list of available Reynolds numbers, then that of the angles of attack.

Note

Unlike the Results tab, which overlays the airfoils, this tab displays **a single operating point at a time**, for a detailed reading of the pressure and the boundary layer.

7.2 Top plot: pressure coefficient

The top plot follows the *XFoil* convention: the vertical axis carries the pressure coefficient C_p **pointing downward** — suction ($C_p < 0$) is therefore directed *upward*. Three curves are overlaid:

- the **upper surface** (yellow);
- the **lower surface** (cyan);
- the **inviscid** C_p (white dashed line), obtained from an XFoil calculation without boundary layer (panel method, Reynolds-independent).

The gap between the viscous and inviscid C_p reveals the effect of the boundary layer: small at low angles of attack, it deepens near the trailing edge (recompression) and in case of separation.

7.3 Bottom plot: airfoil and boundary layer

The bottom plot shows the airfoil **to chord scale** (isometric frame: the airfoil proportions are preserved). Around the surface (in white) are drawn the **boundary layer edges** — the surface offset by the displacement thickness δ^* along the outward normal:

- upper surface in yellow;
- lower surface in cyan.

The thickening of these edges toward the trailing edge reflects the growth of the boundary layer; a marked bulge signals separation. The wake, downstream of the trailing edge, is excluded from the plot.

7.4 Coefficients inset

An inset, at the top right, recalls the integral coefficients at the selected point: Reynolds, angle of attack α, lift C_L, moment C_M, drag C_D and lift-to-drag ratio C_L/C_D.

Tip

Compare the viscous C_p with the inviscid dashed line to assess at a glance the influence of viscosity at the chosen angle of attack; then inspect the δ^* edges to spot separation starting at the trailing edge.

Note

The inviscid C_p is computed by a separate XFoil pass (without boundary layer). This pass is very fast and can be disabled via the `INVISCID_CP` parameter (enabled by default).

8

File formats

AirfoilTools can read and write several airfoil formats, which fall into two families: the *discrete points* formats (originating from the aeronautics field) and the *spline* formats specific to AirfoilTools.

8.1 Discrete points formats

8.1.1 Selig format (`.dat`)

The most widespread historical format, popularized by the University of Illinois (UIUC) database. It is the default format on `airfoiltools.com`.

- First line: airfoil name (free).
- Following lines: `x y` pairs separated by spaces.
- Traversal in *Selig convention*: from the trailing edge ($x = 1$) towards the upper surface (decreasing x), to the leading edge ($x = 0$), then towards the lower surface (increasing x) and back to the trailing edge.

```
NACA 2412
    1.000000   0.001260
    0.985700   0.003930
    0.971440   0.006580
    ...
    0.014300   0.018780
    0.000000   0.000000
    0.014300  -0.014050
    ...
    0.985700  -0.001280
    1.000000  -0.001260
```

Listing 8.1: Example Selig file (NACA 2412)

8.1.2 Lednicer format (`.dat`)

A variant of Selig where upper surface and lower surface are stored separately, each traversed from the leading edge towards the trailing edge. A header line indicates the number of points on each side.

```
NACA 2412
65.0   65.0

    0.000000   0.000000
    0.001425   0.005700
    ...
```

```
    1.000000    0.001260

    0.000000    0.000000
    0.001425   -0.005720
    ...
    1.000000   -0.001260
```

Listing 8.2: Example Lednicer file

8.1.3 CSV format (`.csv`)

A tabular format with header `x,y`. Convenient for importing into spreadsheets or for sharing with non-aeronautical tools.

```
x,y
1.000000,0.001260
0.985700,0.003930
...
```

Listing 8.3: Example CSV file

8.1.4 GNU format (`.gnu`)

A raw points format compatible with the **Axile** software and directly usable by **gnuplot**. It is a simple text file of two columns `x y`, without header or airfoil name.

- One line per point, two columns `x y`.
- Each number in scientific notation with 18 decimals (`%.18e`), separated by a single space.
- `LF` line endings (including after the last point), guaranteed even on Windows.
- No header, comment or third column.

```
1.000000000000000000e+00 0.000000000000000000e+00
9.754952615109063752e-01 6.041608209341215764e-03
0.000000000000000000e+00 0.000000000000000000e+00
```

Listing 8.4: Example GNU file (.gnu)

This format is produced identically by `numpy.savetxt` and readable back by `numpy.loadtxt` (array of shape $(n, 2)$).

Note

The `.gnu` format is offered as **write-only** (save of the current airfoil). To re-import points, use a `.dat` (Selig) or `.csv` file.

8.2 AirfoilTools spline formats

8.2.1 `.bspl` format

A complete save of an airfoil in **multi-segment spline** mode. It preserves the entire Bézier definition: control points, degrees, continuities at junctions, sampling parameters.

This is the **recommended** format to keep the editing work on an airfoil and re-open it without loss of information.

Note

The `.bspl` format is a structured text format (JSON or similar) specific to AirfoilTools. It is not interoperable with other CAD or aerodynamics software. For external distribution, re-export to `.dat` (Selig) after adjusting the sampling.

8.2.2 `.bez` format

A simplified variant of the `.bspl` for airfoils defined by a single Bézier curve per side (single-segment mode). Kept for compatibility with earlier versions.

8.3 Project format (`.aftproj`)

The `.aftproj` format saves the **entire working session** in a single file, where the preceding formats only save an isolated airfoil. A project contains:

- the **current airfoil** and the **reference airfoil** (with their complete spline definition where applicable);
- the optional **tracing image**, *embedded* in the file, together with its **transformation** (position, scale, rotation, visibility).

It is a **JSON** file (text). The image is encoded in base64 and embedded directly: the project is therefore **self-contained and portable**. The tracing image is recovered on reopening even if the original image file has been moved or deleted.

- Open: File → Open project... (Ctrl+Maj+P).
- Save: File → Save project... (Ctrl+Alt+S).

Important

The `.aftproj` format is the **only** one that keeps the tracing image and its alignment. If your work involves tracing (see section 4.8), save a project rather than a simple airfoil.

Note

Since the image is embedded in base64, a project containing a large image produces an `.aftproj` file larger than the sum of the airfoil files. This is the price of portability (no dependency on an external image file).

8.4 NACA generation

The application can generate NACA airfoils on the fly from their designation, without going through a file:

- NACA 4 digits (e.g. `2412`, `0012`): maximum camber (%), position of the max camber (tenths), relative thickness (%).
- NACA 5 digits (e.g. `23012`): cambered airfoil with reflex mean line.

This generation happens automatically at application startup (NACA 2412 as the current airfoil and as the reference). To generate other NACA airfoils during a session, use the File → NACA airfoil → Current airfoil... or ...Reference airfoil... menu: a dialog asks for the indices, then the number of points. See the detailed procedure in section 4.4, page 17.

Note

A NACA airfoil not yet modified can be cleanly regenerated at another resolution via the Number of points... context menu (right-click), which recalls the analytical formula instead of interpolating the points. See section 4.4.

8.5 Summary table

Format	Read	Write	Preserves the spline?
Selig (`.dat`)	yes	yes	no
Lednicer (`.dat`)	yes	yes	no
CSV (`.csv`)	yes	yes	no
GNU (`.gnu`)	no	yes	no
Spline (`.bspl`)	yes	yes	**yes**
Bézier (`.bez`)	yes	yes	**yes** (single-segment)
Project (`.aftproj`)	yes	yes	**yes** (+ tracing)

Important

When you save an airfoil in `.dat` or `.csv` format, **the spline information is lost**: only the list of sampled discrete points is written. To fully preserve your work, always save in `.bspl`.

9

Appendices

9.1 Keyboard shortcuts

Shortcut	Action
Ctrl + O	Open current airfoil
Ctrl + Shift + O	Open reference airfoil
Ctrl + S	Save current airfoil
Ctrl + Q	Quit
Ctrl + Z	Undo (reserved, not active)
Ctrl + Y	Redo (reserved, not active)
Ctrl + B	Convert to spline
Ctrl + 0	Fit zoom (airfoils canvas)

9.2 Glossary

LE / TE Leading edge ($x = 0$) / Trailing edge ($x = c$).

Camber Maximum height of the airfoil mean line, expressed as a percentage of the chord.

C_D Drag coefficient. Drag force normalized by the dynamic pressure and the chord.

C_f Local wall skin-friction coefficient.

C_L Lift coefficient.

C_M Pitching moment coefficient, generally referenced to the quarter chord.

C_p Pressure coefficient: $(p - p_\infty)/(\frac{1}{2}\rho V_\infty^2)$.

C^0, C^1, C^2 continuity At the junction point of two curves: identity of position (C^0), of tangent (C^1), of curvature (C^2).

Chord Distance between the LE and the TE, used as the reference length.

Bézier curve Parametric curve defined by a control polygon. The degree equals the number of control points minus one.

δ^* Boundary-layer displacement thickness.

De Casteljau Recursive algorithm for evaluating and subdividing a Bézier curve. The subdivision is *exact* (no approximation).

Stall Abrupt loss of lift beyond a critical angle of attack, due to boundary-layer separation.

Upper surface / Lower surface Upper / lower surface of the airfoil.

L/D C_L/C_D ratio. The higher the lift-to-drag ratio, the better the aerodynamic efficiency.

H Boundary-layer shape factor, $H = \delta^*/\theta$.

NCRIT Transition criterion in the e^N method; maximum accepted amplitude of disturbances before the onset of turbulence.

Polar Plot of one aerodynamic quantity as a function of another, at a given Reynolds number. Several polars form a family indexed by the Reynolds number.

Reynolds Dimensionless number $\mathrm{Re} = \rho V c/\mu$, characterizing the flow regime (laminar, transitional, turbulent).

Selig Convention for storing the points of an airfoil: TE → upper surface → LE → lower surface → TE.

Spline Piecewise curve composed of several Bézier segments joined according to a given level of continuity.

θ Boundary-layer momentum thickness.

Trip / Turbulator Element (roughness, stretched wire) forcing the laminar-to-turbulent transition at a given position.

U_e Velocity at the outer edge of the boundary layer.

XFoil 2D aerodynamic solver developed by Mark Drela (MIT) coupling a panel method with boundary-layer integral equations.

XTR Boundary-layer transition position (x_{tr}/c).

9.3 Troubleshooting (FAQ)

The GUI does not launch

Check that:

1. the virtual environment is activated (`env_py3`);
2. the dependencies are installed (`pip install -r requirements.txt`);
3. Python is version 3.10 or higher (`python -version`).

XFoil does not run

- Check for the presence of `externaltools/xfoil/xfoil.exe`.
- On Windows, some antivirus programs may block execution. Add an exception for the `externaltools` folder.
- The commands sent to XFoil are visible in the console (in development mode); review the output to diagnose any syntax error.

XFoil does not converge for some angles

- Disable ASEQ to switch to bidirectional ALFA mode.
- Reduce the α step (for example from 0.5° to 0.25°) to ease continuation tracking.
- Increase Timeout if the computation is long.
- Check that the airfoil has no geometric singularities (duplicated points, upper/lower surface crossing, degenerate LE).

The curvature is not displayed

- Convert the airfoil to spline mode (Edit → Convert to Spline).
- Increase the porcupine scale (right-click on the canvas → Porcupine scale...) if they are too short to be visible.

I lose my splines when saving

The `.dat` format (Selig/Lednicer) does not preserve the spline representation: only the discrete points are saved. To keep the spline, use the `.bspl` format.

9.4 Further reading

- **Technical documentation**: see the docstrings in the source code (Sphinx reST format) and the documentation generated in the `docs/` folder.
- **Unit tests**: see the `tests/` folder, which contains more than 240 tests (Bezier, Profil, BezierSpline, ProfilSpline, Simulation).
- **Reference article on XFoil**: Drela, M. *XFOIL: An Analysis and Design System for Low Reynolds Number Airfoils*, in *Low Reynolds Number Aerodynamics*, Springer Lecture Notes in Engineering, Vol. 54, 1989.
- **Airfoil library**: `https://m-selig.ae.illinois.edu/ads/coord_database.html` (large catalogue in `.dat` format, accessible directly from the application's File → From the UIUC database menu).

End of the user manual.
To report a typo, suggest an improvement or ask a question, open an issue *on* *`https://github.com/BenoitGFree/AirfoilTools`.*